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Emotionally Aware Music Playback : Mood Sync

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “Emotionally Aware Music Playback : Mood Sync” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Mrs. Simarjit Virk. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Abstract

In the modern era, rapidly evolving lifestyles and continuous digital exposure have led to a significant rise in stress, anxiety, and emotional exhaustion across individuals of all age groups, making mental health management an essential concern. Among various therapeutic approaches, music therapy has emerged as an effective and accessible method for regulating emotions and improving overall well-being, as evidenced by a recent study involving 256 participants that reported a noticeable enhancement in emotional resilience. Despite its benefits, conventional music playback systems fail to provide personalized experiences, as they rely on manual user input rather than adapting to the user's real-time emotional state. To overcome this limitation, this report introduces MoodSync, an AI-driven system designed to deliver adaptive music therapy by detecting the user's mood through webcam-based facial expression analysis. The system employs a Convolutional Neural Network (CNN) trained on an emotion-annotated dataset to accurately recognize emotional states, which are then integrated with the user's language preferences, including English, Hindi, and Punjabi, to curate suitable music automatically.

Furthermore, the system architecture integrates a robust backend developed using Node.js and Express with an emotion recognition module implemented using Python and Keras libraries, ensuring seamless communication between components and efficient data processing. MoodSync achieved an accuracy of 80% in emotion detection during testing and demonstrated reliable real-time performance, making it suitable for practical deployment. The system is designed to be flexible and scalable, allowing future enhancements such as incorporating additional languages, expanding emotion categories, and integrating wearable sensor data for improved accuracy. Overall, MoodSync not only enhances personalized music experiences but also contributes significantly to the development of intelligent, emotion-aware systems in the fields of mental health support and human-computer interaction.

1. Introduction

1.1. Background

In today's era, rising levels of stress and emotional imbalance are very common among students, working professionals, and the general population. Despite multiple music streaming platforms being available, most of them rely on user intervention and ignore the listener's real-time emotional states. People are expected to manually curate playlists or search for music that matches their mood, which is counterproductive when they are already emotionally distressed. MoodSync aims to bridge this gap by introducing an intelligent system that understands human emotions through facial expressions and responds with contextually appropriate music, thereby providing a hands-free, automated emotional music therapy experience.

1.2. Evolution of Emotion-Aware Systems

Advancements in technology have enabled machines to interpret human emotions through facial recognition, voice analysis, physiological signals, and other modalities. Technologies such as deep learning and computer vision have significantly improved the accuracy of emotion recognition. Research in affective computing has increasingly focused on using these technologies for real-world applications including healthcare, education, entertainment, and customer service.

1.3. Importance in Modern Education and Well-being

In this fast-paced world, mental health awareness has become crucial among students and working professionals. To help combat this situation, MoodSync incubates music therapy - a non-invasive method to regulate emotions. It is an automated system that plays music according to the user's mood and language preference, making it particularly relevant for students experiencing academic stress and professionals dealing with workplace pressures.

1.4. Purpose and Scope of the Study

The objective of this research is to combat the shortcomings of conventional music systems, which rely on user intervention. Such approaches generally fail to incorporate real-time emotional awareness and cannot adapt dynamically to the listener's present emotional state. The primary scope of this study is to bridge this gap by incorporating emotion recognition

techniques with music playback mechanisms, thereby creating a more personalized music playback system. This work contributes to the development of affective computing applications in entertainment and well-being by enabling systems that respond sensitively to users' immediate affective needs rather than static profiles alone.

1.5.Objectives

- To develop a real-time facial emotion detection system using Convolutional Neural Networks (CNN).
- To integrate emotion detection with a music playback system that supports multiple languages (English, Hindi, Punjabi).
- To create a responsive and user-friendly web interface for the MoodSync system.
- To evaluate the performance of the emotion detection model in real-world conditions.
- To explore the therapeutic potential of emotion-aware music playback for mental well-being.

2. Methodology

2.1. Research Approach

The research methodology for MoodSync follows a systematic and structured approach. It begins with a comprehensive literature review of existing emotion detection and music recommendation systems, followed by dataset selection, model development, system integration, and final evaluation. Each stage was carefully planned to ensure the robustness and effectiveness of the proposed system.

2.2.System Architecture

The MoodSync system incorporates a technical architecture designed with flexibility, maintainability, and seamless integration across its core components. The system consists of four main modules:

- **Emotion Detection Module:** Built in Python, this module uses OpenCV for real-time face detection and preprocessing, paired with a CNN model to accurately identify the user's current emotional state from facial expressions.

- **Music Playback Module:** Maps the detected emotion to an appropriate music playlist based on the user's language preference.
- **Backend Module:** Built using Node.js and Express, it manages communication between the frontend and the emotion detection component.
- **User Interface Module:** A responsive web-based platform that enables webcam access and displays real-time emotion feedback to the user.

2.3.Data Preparation

The system uses emotion-labeled datasets to train the CNN model. The dataset used in this project contains 35,685 facial emotion-recognized images, where each image is resized to 48x48 pixels. The resized images are subsequently converted into grayscale format to reduce computational complexity while preserving all key facial features. The data is divided into training and testing subsets. Training data is used by the model to learn patterns, while testing data is used to evaluate the model's performance on unseen data. The images are grouped into emotion-specific folders; for instance, the 'angry' emotion category contains approximately 958 image files. The dataset exhibits class imbalance, with emotions like 'happy' and 'neutral' having more image samples than others.

2.4.Data Analysis

Music generation in MoodSync uses the detected emotion and the user's language preference (English, Hindi, or Punjabi) to build a dynamic mood-based music delivery system. After data collection, the data is organized and prepared for analysis. Noisy or inconsistent data is removed to improve the accuracy and reliability of the system.

2.5.Evaluation of Key Factors

The collected data is analyzed to evaluate key factors affecting MoodSync performance, including emotion detection accuracy and reliability, input image quality, music playback effectiveness, and real-time responsiveness. This evaluation helps assess how well the system adapts to the user's emotional state, delivers personalized mood-based music, and supports emotional well-being.

3. Tools and Technologies

3.1. Programming Languages

- Python: Used for developing the emotion recognition component, including data preprocessing, CNN model training, and real-time inference using OpenCV.
- JavaScript (Node.js): Used for building the backend server and managing API communication between system components.
- HTML/CSS: Used for developing the frontend user interface.

3.2. Frameworks and Libraries

- Keras: A high-level deep learning API used to build and train the Convolutional Neural Network (CNN) model for emotion classification.
- TensorFlow: Serves as the backend engine for Keras, powering the deep learning computations.
- OpenCV: An open-source computer vision library used for real-time face detection and image preprocessing from webcam input.
- Express.js: A minimal Node.js web framework used to build the backend REST API and manage server-side operations.

3.3. Dataset

The FER (Facial Emotion Recognition) dataset sourced from Kaggle (<https://www.kaggle.com/datasets/ananthu017/emotion-detection-fer>) was used for training and testing the CNN model. The dataset consists of 35,685 grayscale images of size 48x48 pixels, categorized into different emotion classes including Happy, Sad, Angry, Neutral, Surprise, and others.

3.4. Development Tools

- Visual Studio Code: Primary Integrated Development Environment (IDE) used for writing and debugging code.
- Jupyter Notebook: Used for data exploration, model training experiments, and evaluation.
- Postman: Used for testing and verifying backend API endpoints.
- GitHub: Used for version control and collaboration.

3.5. Platform and Deployment

The system runs on a local server environment, with the Node.js backend serving the frontend application. The emotion detection component runs as a Python subprocess communicating with the Node.js server. The webcam input is captured via the browser's MediaDevices API on the client side.

4. Implementation

4.1. System Overview

The implementation of MoodSync involves a multi-layer architecture combining computer vision, deep learning, and web technologies. The system flow begins when the user accesses the web interface, selects a preferred language, and activates the webcam. The emotion detection module then processes the video feed in real-time to identify the user's emotional state, which is subsequently used to trigger mood-appropriate music playback.

4.2. Emotion Detection Module

The emotion detection module was developed in Python using OpenCV and Keras. The implementation steps are as follows:

- **Face Detection:** OpenCV's Haar Cascade Classifier is used to detect faces in real-time webcam frames.
- **Preprocessing:** Detected face regions are extracted, resized to 48x48 pixels, and converted to grayscale to match the training data format.
- **Emotion Classification:** The preprocessed image is passed to the trained CNN model, which outputs a probability distribution across emotion categories. The emotion with the highest probability is selected as the detected emotion.

4.3. CNN Model Architecture

The Convolutional Neural Network (CNN) model was built using Keras with the following key components:

- **Convolutional Layers:** Multiple convolutional layers with ReLU activation functions to extract spatial features from facial images.
- **Max Pooling Layers:** Used after convolutional layers to reduce spatial dimensions and computational complexity while retaining key features.

- Dropout Layers: Applied for regularization to prevent overfitting during training.
- Fully Connected Layers: Dense layers to classify extracted features into one of the emotion categories.
- Output Layer: A softmax activation function to produce a probability distribution across emotion classes.

4.4.Backend and API Integration

The Node.js and Express backend serves as the communication bridge between the frontend user interface and the Python emotion detection module. When the frontend captures a webcam frame, it sends it to the backend via an API call. The backend triggers the Python script to process the image, receives the detected emotion, and returns it to the frontend for display. The backend also manages the music playback logic based on the detected emotion and the user's selected language.

4.5.Frontend Interface

The frontend is built using HTML, CSS, and JavaScript. The interface provides the user with a language selection dropdown (English, Hindi, Punjabi), a webcam preview window, a real-time emotion display panel, and a music player. The webcam is accessed using the browser's MediaDevices API, and frames are periodically sent to the backend for emotion analysis.

4.6.Music Playback Logic

MoodSync uses a rule-based music mapping system. Based on the detected emotion and user-selected language, the system selects and plays a corresponding music track from a pre-organized music library. The mapping rules are as follows:

- Happy Emotion: Upbeat and energetic music tracks.
- Sad Emotion: Comforting and uplifting music to console the user.
- Angry Emotion: Calm and soothing music to reduce agitation.
- Neutral Emotion: Ambient or light background music.
- Surprise/Calm Emotion: Relaxing and melodic tracks.

5. Major Findings

5.1. System Testing

The proposed MoodSync system was tested in real time to evaluate its performance. The system was tested across a sample of 10 real-time detections under varied conditions including different lighting environments, facial angles, and expressions. The observations confirm that the system was able to accurately detect emotions and play context-relevant music in most tested scenarios.

5.2. Emotion Detection Performance

During testing, the system successfully identified the following emotions: Happiness, Sadness, Anger, Calmness, Surprise, and Neutrality. The system demonstrated an overall accuracy of 80% based on a sample of 10 detections, with 8 out of 10 samples correctly classified.

The following results were observed in sample detections:

- Detection of Happy emotion led to the playback of upbeat, energetic music.
- Detection of Sad emotion led to the playback of comforting and joyful music to console the user.
- Detection of Angry emotion led to the playback of calming and soothing music.

These results demonstrate the capability of the system to accurately map detected emotions to contextually appropriate music.

5.3. Confusion Matrix Analysis

The following confusion matrix (Table 1.1) presents a detailed breakdown of the system's prediction accuracy across four emotion classes based on 10 test samples:

Actual Emotion	Pred. HAPPY	Pred. SAD	Pred. ANGRY	Pred. NEUTRAL
HAPPY	2	0	0	0

SAD	0	2	0	0
ANGRY	0	1	1	0
NEUTRAL	0	0	1	1
TOTAL	2	3	2	1

Table 1.1: Confusion Matrix depicting the accuracy of the proposed model.

5.4. Real-Time Performance

The system demonstrated efficient real-time performance, with a measured latency of approximately 45 seconds between the initial emotion detection trigger and the commencement of music playback. This response time is considered acceptable for the intended use-case of ambient music therapy, where immediate playback is less critical than accuracy of emotion mapping.

5.5. Key Observations

- The system performed best under well-lit conditions with the user's face clearly visible and centred in the webcam frame.
- Performance degraded under poor lighting conditions or when the face was partially obstructed or turned at extreme angles.
- The multilingual support feature (English, Hindi, Punjabi) was found to enhance user engagement and inclusivity.

The system's real-time emotion display provided useful feedback to users, increasing their trust and interaction with the system.

6. Conclusion and Future Scope

6.1. Conclusion

In conclusion, this report presents MoodSync — an emotionally aware music playback system that complements current advancements in the field of Artificial Intelligence. MoodSync has been explored as a potential supplement in the domain of mental health and well-being, capable of utilizing music therapy for emotional regulation. The system traversed through various critical factors influencing its effectiveness, including the strength of the CNN model, the variety in the input dataset, and the focus on user satisfaction. The results of this research indicate that MoodSync was able to effectively identify emotions with 80% accuracy and play context-relevant music, which highlights

MoodSync's capability to positively influence the user's mood. The system's ability to support multiple languages further enhances its accessibility and therapeutic relevance across diverse user groups. However, the system remains sensitive to factors such as lighting conditions, image quality, and facial alignment, which can lead to unexpected results. Despite these challenges, MoodSync represents a pragmatic and constructive solution in the discipline of mental health and human-computer interaction.

6.2.Future Scope

Several directions for future work have been identified to enhance the capabilities and robustness of MoodSync:

- **Advanced Model Training:** Training the CNN model on larger and more diverse datasets to improve generalization and accuracy across varied demographic groups and real-world conditions.
- **Multimodal Emotion Detection:** Integrating additional emotion modalities such as voice analysis, physiological signals (heart rate, skin conductance), and body language for more accurate and robust emotion detection.
- **Expanded Language and Genre Support:** Extending the music library to include more regional languages and a wider variety of music genres to cater to a broader audience.
- **Mobile Application Development:** Developing a dedicated mobile application to make MoodSync more accessible and portable for everyday use.
- **Longitudinal Studies:** Conducting longitudinal studies to evaluate the long-term therapeutic impact of MoodSync on users' mental health and emotional well-being.
- **Personalization through Machine Learning:** Incorporating user feedback and listening history to build personalized emotion-to-music mapping models using reinforcement learning techniques.
- **Cloud Deployment:** Deploying the system on cloud infrastructure for scalability and wider accessibility beyond local network environments.

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